

Computer Networks Spring 2026

Assignment#3 (6A & 6B) Solution

Part – 1

Review Questions:

R6.

6. Yes. The application developer can put reliable data transfer into the application layer protocol. This would require a significant amount of work and debugging, however.

R15.

15. a) 20 bytes b) ack number = 90

Problems:

Problem 2

Suppose the IP addresses of the hosts A, B, and C are a, b, c, respectively. (Note that a, b, c are distinct.)

To host A: Source port = 80, source IP address = b, dest port = 26145, dest IP address = a

To host C, left process: Source port = 80, source IP address = b, dest port = 7532, dest IP address = c

To host C, right process: Source port = 80, source IP address = b, dest port = 26145, dest IP address = c

Problem 3

Note, wrap around if overflow.

$$\begin{array}{r} 0 \ 1 \ 0 \ 1 \ 0 \ 0 \ 1 \ 1 \\ + \ 0 \ 1 \ 1 \ 0 \ 0 \ 1 \ 1 \ 0 \\ \hline 1 \ 0 \ 1 \ 1 \ 1 \ 0 \ 0 \ 1 \end{array}$$

$$\begin{array}{r}
 1\ 0\ 1\ 1\ 1\ 0\ 0\ 1 \\
 +\ 0\ 1\ 1\ 1\ 0\ 1\ 0\ 0 \\
 \hline
 0\ 0\ 1\ 0\ 1\ 1\ 1\ 0
 \end{array}$$

One's complement = 1 1 0 1 0 0 0 1.

To detect errors, the receiver adds the four words (the three original words and the checksum). If the sum contains a zero, the receiver knows there has been an error. All one-bit errors will be detected, but two-bit errors can be undetected (e.g., if the last digit of the first word is converted to a 0 and the last digit of the second word is converted to a 1).

Problem 31

$$\text{DevRTT} = (1 - \beta) * \text{DevRTT} + \beta * |\text{SampleRTT} - \text{EstimatedRTT}|$$

$$\text{EstimatedRTT} = (1 - \alpha) * \text{EstimatedRTT} + \alpha * \text{SampleRTT}$$

$$\text{TimeoutInterval} = \text{EstimatedRTT} + 4 * \text{DevRTT}$$

After obtaining first SampleRTT 106ms:

$$\text{DevRTT} = 0.75 * 5 + 0.25 * |106 - 100| = 5.25 \text{ ms}$$

$$\text{EstimatedRTT} = 0.875 * 100 + 0.125 * 106 = 100.75 \text{ ms}$$

$$\text{TimeoutInterval} = 100.75 + 4 * 5.25 = 121.75 \text{ ms}$$

After obtaining 120ms:

$$\text{DevRTT} = 0.75 * 5.25 + 0.25 * |120 - 100.75| = 8.75 \text{ ms}$$

$$\text{EstimatedRTT} = 0.875 * 100.75 + 0.125 * 120 = 103.16 \text{ ms}$$

$$\text{TimeoutInterval} = 103.16 + 4 * 8.75 = 138.16 \text{ ms}$$

After obtaining 140ms:

$$\text{DevRTT} = 0.75 * 8.75 + 0.25 * |140 - 103.16| = 15.77 \text{ ms}$$

$$\text{EstimatedRTT} = 0.875 * 103.16 + 0.125 * 140 = 107.76 \text{ ms}$$

$$\text{TimeoutInterval} = 107.76 + 4 * 15.77 = 170.84 \text{ ms}$$

After obtaining 90ms:

$$\text{DevRTT} = 0.75 * 15.77 + 0.25 * |90 - 107.76| = 16.27 \text{ ms}$$

$$\text{EstimatedRTT} = 0.875 * 107.76 + 0.125 * 90 = 105.54 \text{ ms}$$

$$\text{TimeoutInterval} = 105.54 + 4 * 16.27 = 170.62 \text{ ms}$$

After obtaining 115ms:

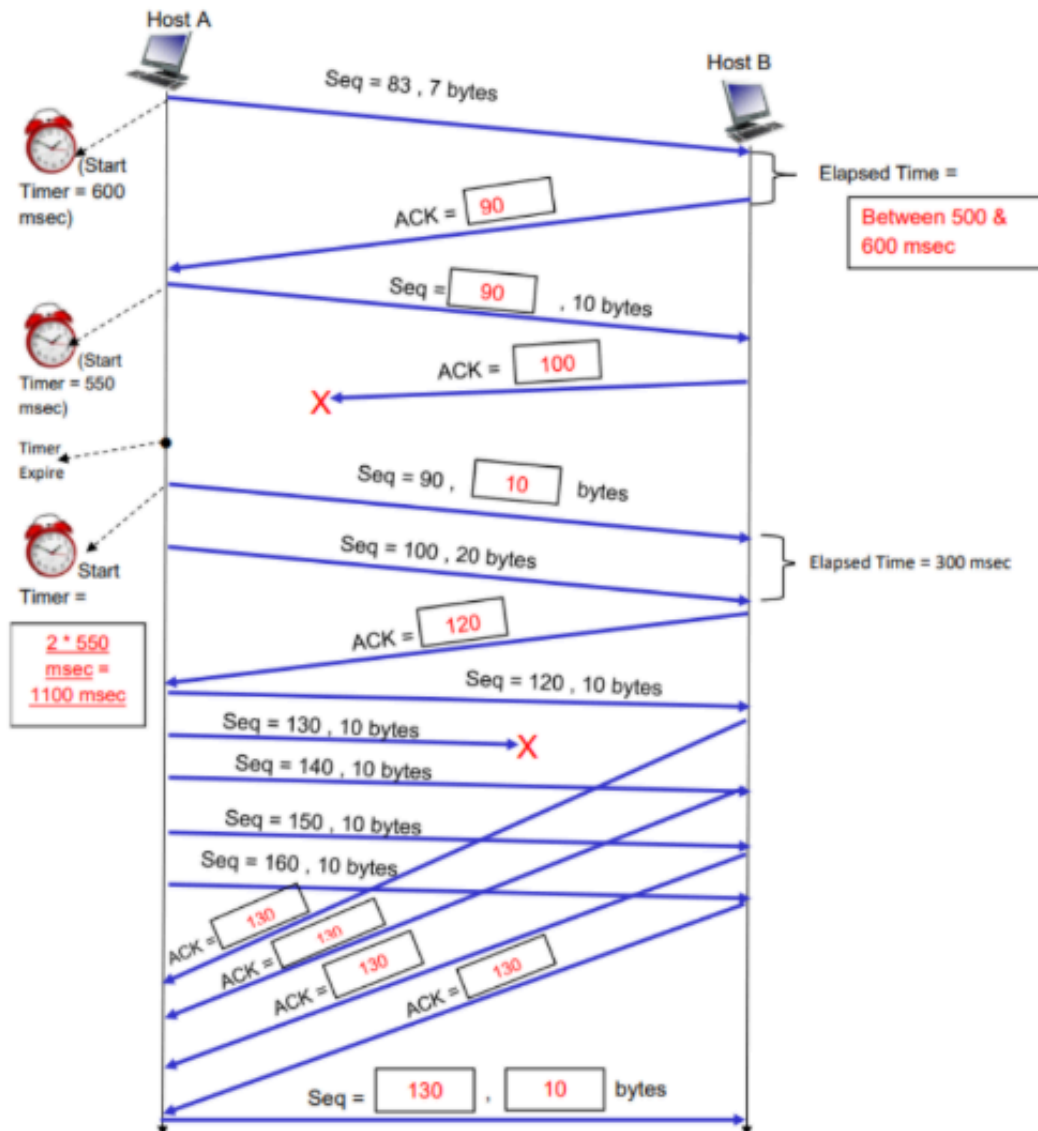
$$\text{DevRTT} = 0.75 * 16.27 + 0.25 * |115 - 105.54| = 14.57 \text{ ms}$$

$$\text{EstimatedRTT} = 0.875 * 105.54 + 0.125 * 115 = 106.72 \text{ ms}$$

$$\text{TimeoutInterval} = 106.72 + 4 * 14.57 = 165 \text{ ms}$$

Part – 2

Question: 01



Question 2:

Triggered by 3 duplicate ACKs (4 identical ACKs total). If the sender receives 3 dup-ACKs for sequence X, it assumes X+1 was lost (since the receiver is repeatedly asking for it despite getting later segments) and retransmits immediately before the timer expires.

Question 3:

Errors can occur inside routers (memory corruption) or during protocol translation where Link Layer CRC is stripped off and recalculated. The End-to-End principle states that reliability must be verified by the application/transport endpoints, as lower-level checks are insufficient to guarantee data integrity across the entire path.

Question 4:

- i. In Selective Repeat, only the lost packets are retransmitted. The sender retransmits only packet 5 and packet 9. Total 2 packets are retransmitted.
- ii. In Go-Back-N, if a packet is lost, the sender retransmits the lost packet and all subsequent packets in the window. Since packet 6 is lost. The sender retransmits packet 6, 7, 8, 9, and 10. Total 5 packets are retransmitted in total.

Question 5:

- b. What is the sequence number of the segment which was lost? 1300
- a. Complete Table on next page

Solution

TXN Round (RTT)	Receive Window (rwnd in MSS)	Congestion Window (cwnd in MSS)	Slow Start Threshold (ssthresh in MSS)	Received ACK Number	Sequence Number(s) of Transmitted Segment(s)	Timer (Timeout in ms)	No Loss Or Timeout Or 3dupAck	TCP Sender Congestion Mode (SS or CA or FR)
0	6	0	12	N/A	N/A	600	No Loss	N/A
1	6	<u>1</u>	<u>12</u>	N/A	<u>0</u>	600	No Loss	SS
2	6	<u>2</u>	<u>12</u>	<u>100</u>	<u>100 , 200</u>	600	<u>No Loss</u>	<u>SS</u>
4	6	8	<u>12</u>	<u>700</u>	<u>700 , 800 , 900 , 1000 , 1100 , 1200</u>	600	No Loss	<u>SS</u>
5	6	<u>12</u>	12	1300	<u>1300 , 1400 , 1500 , 1600 , 1700 , 1800</u>	600	<u>No Loss</u>	<u>CA</u>
6	20	<u>1</u>	<u>6</u>	<u>1300</u>	<u>1300</u>	<u>1200</u>	<u>Timeout</u>	<u>SS</u>
10	20	<u>7</u>	<u>6</u>	<u>2600</u>	<u>2600 , 2700 , 2800 , 2900 , 3000 , 3100 , 3200</u>	1200	<u>No Loss</u>	<u>CA</u>

Complete Table (Including missing rows)

TXN Round (RTT)	Receive Window (rwnd in MSS)	Congestion Window (cwnd in MSS)	Slow Start Threshold (ssthresh in MSS)	Received ACK Number	Sequence Number(s) of Transmitted Segment(s)	Timer (Timeout in ms)	No Loss Or Timeout Or 3dupAck	TCP Sender Congestion Mode (SS or CA or FR)
0	6	0	12	N/A	N/A	600	No Loss	N/A
1	6	1	12	N/A	0	600	No Loss	SS
2	6	2	12	100	100 , 200	600	No Loss	SS
3	6	4	12	300	300 , 400 , 500, 600	600	No Loss	SS
4	6	8	12	700	700 , 800 , 900 , 1000 , 1100 , 1200	600	No Loss	SS
5	6	12	12	1300	<u>1300</u> , 1400 , 1500 , 1600 , 1700 , 1800	600	No Loss	CA
6	20	1	6	1300	1300	1200	Timeout	SS
7	20	2	6	1400	1400 , 1500	1200	No Loss	SS
8	20	4	6	1600	1600 , 1700 , 1800 , 1900	1200	No Loss	SS
9	20	6	6	2000	2000 , 2100 , 2200 , 2300 , 2400 , 2500	1200	No Loss	CA
10	20	7	6	2600	2600 , 2700 , 2800 , 2900 , 3000 , 3100 , 3200	1200	No Loss	CA